Large-Scale Pan-Cancer Transcriptomic Analysis Using TCGA Data

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Introduction

Cancer research has historically been challenging due to the high heterogeneity observed across different cancer types. Previous pan-cancer studies have shown that while most tumors do not share the same mutations, they often converge onto common downstream signaling pathways. [1]

In this study, we perform a large-scale transcriptomic analysis using TCGA Pan-Cancer RNA-Seq data to identify differentially expressed genes and enriched pathways across multiple cancer types. At the whole-dataset level, this allows us to detect consistently over-expressed genes and pathways, which may represent oncogenes and oncogenic mechanisms, as well as under-expressed genes and pathways, which may correspond to tumor suppressor genes and associated regulatory processes.

We aim to identify differential gene expression patterns and dysregulated pathways that distinguish one cancer type from another. These findings are compared with existing literature to validate the pathways observed in the analysis and to identify potential biomarkers.

A further objective of this study is to analyze the potential metastatic routes tumors can take. The aim is to identify possible metastatic routes, as well as to determine how likely clinical diagnoses can be incorrect, when based on genes and biomarkers present in tissue samples. We will study how well-differentiated each tumor type is.

Dataset Description

The dataset used in this project was obtained from The Cancer Genome Atlas Research Network Pan-Cancer Atlas initiative. The TCGA Pan-Cancer Atlas contains large-scale molecular profiling data across 33 cancer types, including RNA-Seq gene expression data, DNA methylation profiles, clinical metadata, survival metadata, and other multi-omics features.

For this project, the focus was specifically on transcriptomic analysis using the TCGA batch-corrected, RSEM-normalized RNA-Seq gene expression dataset. This dataset is structured as a gene expression matrix containing 20,531 genes and over 10,000 patient samples across 33 different cancer types. Each row corresponds to a gene, while each column corresponds to a TCGA patient sample. The associated clinical metadata provides information such as patient ID, cancer type, gender, vital status, histological type, and pathological stage.

Dataset Preprocessing

The dataset was filtered to retain only cancer types with at least 50 patient samples, in order to improve statistical reliability. After this filtering step, 16 cancer types were selected for further analysis, resulting in a final dataset of 6,053 patient samples.

The selected cancer types were:

BRCA, COAD, ESCA, GBM, KIRC, LGG, LUAD, LUSC, OV, PRAD, READ, TGCT, THCA, THYM, UVM, and CESC.

The RNA-Seq sample identifiers were given in TCGA barcode format. In each barcode, the first three sections represented the unique patient ID, which was checked against the clinical metadata. The fourth section of the barcode was used to identify the sample type. In TCGA barcodes, 01 represents a primary tumor sample, while 11 represents a normal tissue sample. Samples labelled 01 were used for the analysis of the corresponding cancer type, while samples labelled 11 were used as the control/normal tissue group.

Methodology

Dimensionality Reduction and Visualization

PCA, t-SNE and UMAP were performed as part of initial exploratory analyses to visualize the high-dimensional dataset and identify the clusters formed by different cancer types considered under this study.

t-SNE was performed with both relatively low and high values of Perplexity, and UMAP was performed with both relatively low and high values of neighbors as well as high and low number of Principal Components to identify local and global-scale characteristics of the dataset.

Differential Gene Expression (DGE) Analysis

DGE was performed using the limma-trend workflow in R upon performing a log2 transformation on the previously normalized data.

$|log_2FC| > 1$ and **adjusted p-value < 0.05** were considered as inclusion criteria to determine differentially expressed genes and plot the relevant volcano plots.

Pathway Enrichment Analysis (PEA)

PEA was performed using DAVID (using the gene labels for *Homo sapiens*) for the differentially expressed genes as obtained above. [2] The following comparisons were considered and significantly upregulated and downregulated KEGG pathways (**Counts ≥ 5, FDR < 0.05**) were reported in each case:

1.Whole control vs Whole tumor samples

2.Tumors from the Central Cluster (as observed in Figure 1) versus the corresponding Controls

3.Individual comparisons between THCA, KIRC, LGG and GBM, COAD and READ, PRAD (which represent individual clusters) and the Central Cluster

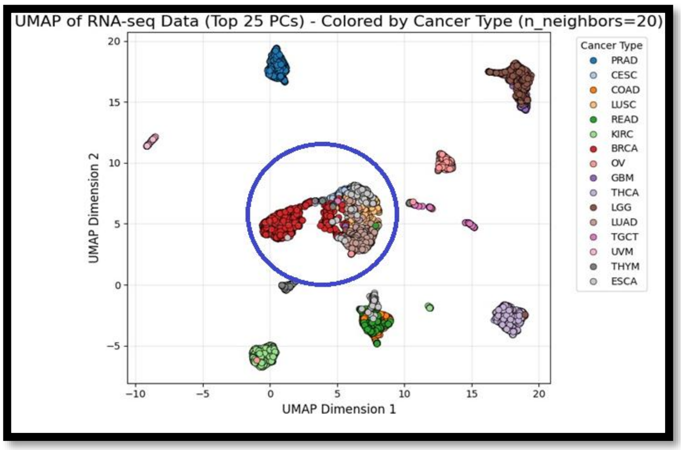

Figure 1

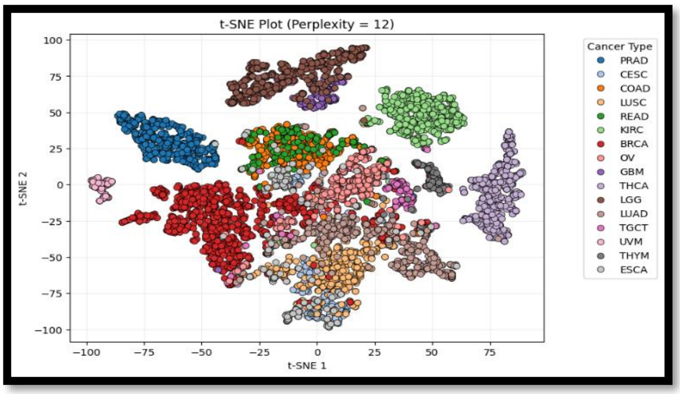

Figure 2

Gaussian Naïve Bayes Classification

The probabilistic, generative Gaussian Naïve Bayes model will show the spread of data points within a given label, and how close their decision boundaries are to the other cancer.

PCA was used to reduce the number of genes/features to 769 (80% of the variance), making it feasible to run the algorithm. Then, the mean vector and covariance matrix for each label were calculated and the latter was compared across the cancer types. The traces, determinants and eigenvalues are an indication of their distribution and roundedness in the feature space.

The parameters obtained were used to predict each sample, and create a confusion matrix. This will show how likely each cancer type is to be misidentified as another.

The number of principal components is then varied. A higher number of components earlier gave a detailed description of the data distribution. Reducing the variance will result in lower accuracy, and thus more susceptibility towards inaccuracies. The trends observed here will be a clearer picture of tumor similarity.

Results and Discussion

Cancer cells vs Normal cells (Overall)

The most prominent features in the upregulated pathways (Fig. 3) are related to proliferation and managing genomic instability. The Cell Cycle pathway was the most enriched pathway. This result correlates well with one of the hallmarks of cancer, where CDK-cyclin complexes are widely overexpressed, allowing for uncontrolled division. [3]

This conclusion is corroborated by the enrichment of the DNA Replication, Fanconi Anemia, and p53 Signaling pathways: tumor cells are actively replicating, experience replication stress, and use their DNA damage response pathways in a modified or bypassed manner. [4]

The enrichment of cellular senescence pathways suggests that some cancers, such as GBM and LGG, may not rely primarily on a hyperproliferative profile, but instead may persist in a more dormant or slow-cycling state for long periods. It might also be that the phenomenon known as oncogene-induced senescence bypass in cancer cells is occurring in some of our cancer types. [3, 4]

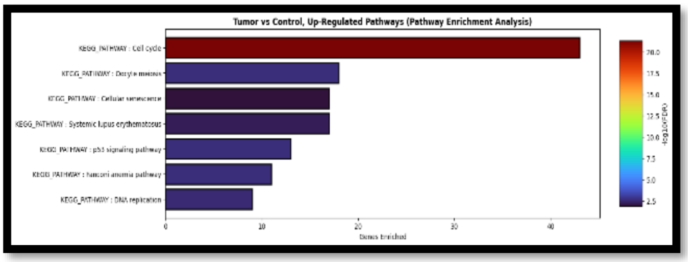

Figure 3

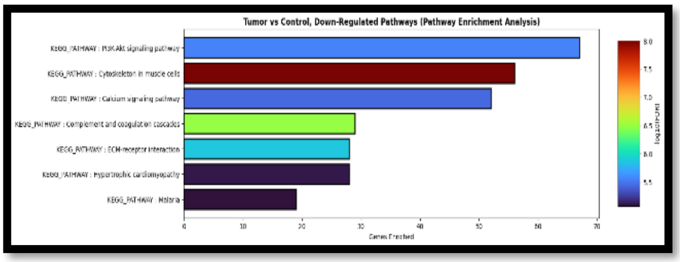

Figure 4

In contrast to the upregulated pathways, the downregulated ones (Fig. 4) demonstrate dysfunctionality in signaling pathways involved in cellular homeostasis and structure.

The PI3K-Akt Signaling pathway, though considered to drive oncogenesis, is downregulated; however, this is due to the downregulation of receptors such as integrins that mediate interactions between the cell and the ECM, a phenomenon which has been observed elsewhere as well. [5]

The loss of the pathways of Calcium Signaling, ECM-receptor interaction and Cytoskeleton in Muscle Cells is another pointer in the same direction: all tissue architecture programs are attenuated, including adhesion and mechano-transduction. [5, 6]

From the above findings, a common pan-tumor signature comprising a general upregulation of cell replication and proliferation pathways and down regulation of DNA repair and apoptotic pathways is seen.

<u>Cancer cells vs Control cells (Central Cluster)</u>

The central cluster showed strong upregulation of cell cycle, mitosis, and DNA replication pathways (Fig. 5). This indicates that the tumor samples are highly proliferative as compared to the control samples. Therefore, the separation of this cluster is mainly driven by a broad tumor-state expression profile rather than a single marker. [7, 8]

The upregulated pathways also included extracellular matrix remodeling. This suggests increased interaction between tumor cells and the surrounding tissue environment, indicating a more invasive phenotype. However, this does not prove metastasis by itself, but supports the possibility that these tumor cells may be better adapted to changing tissue environments. [9]

The downregulated pathways (Fig. 6) are mainly associated with cell-ECM interactions, indicating that these cancer cells have a more invasive phenotype and are more likely to metastasize.

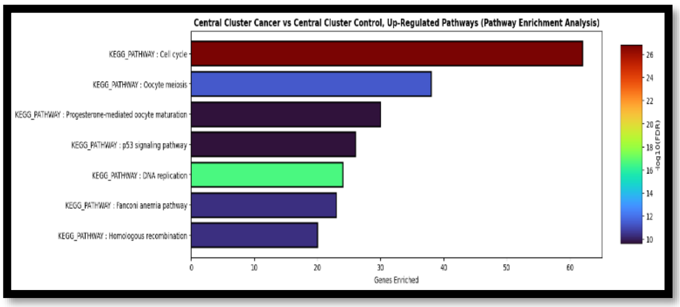

Figure 5

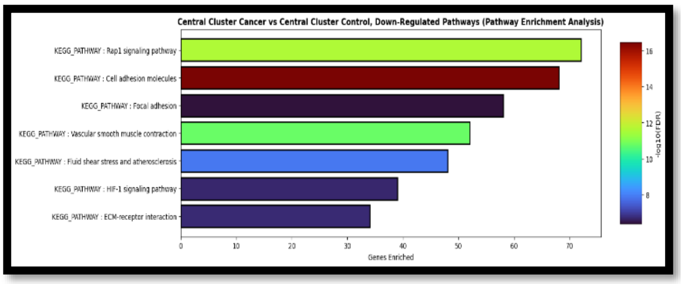

Figure 6

Overall, the central cluster appears to separate from the corresponding control group in two key aspects: gain of proliferative capacity, and loss of cell-ECM interactions. This central cluster consists mainly of carcinoma samples, which is observed in both the upregulated and downregulated pathways, which reflect what is typically seen in carcinomas.

We therefore assume that the central carcinoma cluster serves as a reasonable representation of a “general” cancer. To identify what distinguishes the other cancer clusters seen in Figure 1, each cluster is compared against this central carcinoma cluster in the following sections.

<u>KIRC vs Central Cluster</u>

The upregulated pathways in KIRC include those associated with cell adhesion molecules, focal adhesions, and ECM-receptor interactions (Fig.7), indicating increased interaction between tumor cells and the extracellular matrix. This is consistent with clear cell renal cell carcinoma behavior, where the ECM environment influences key tumor properties such as proliferation, vascularization, and invasion. [10]

The tumor suppressor VHL is lost in over 90% of KIRC cases through genetic or epigenetic mechanisms. Since VHL is central to cellular oxygen sensing, its inactivation leads to persistent pseudo-hypoxia. This is reflected in the upregulation of HIF-1 signaling, along with vascular-associated pathways such as apelin signaling, contributing to the strong angiogenic profile of these tumors. [10]

Additional differences were observed in pathways associated with kidney-specific functions, including renin secretion, drug metabolism, and smooth muscle contraction. These differences are not key indicators, but they reflect the renal origin of the tumor compared with the central carcinoma cluster.

The downregulated pathways mainly include cell cycle, DNA replication, and homologous recombination and p53 signaling pathway (Fig. 8). This indicates that proliferation and repair associated pathways are relatively subdued in KIRC compared with the central carcinoma cluster. Overall, this analysis suggests that clear cell renal carcinoma has a stronger tendency to metastasize over the hypertrophic pathways seen in central carcinoma cluster.

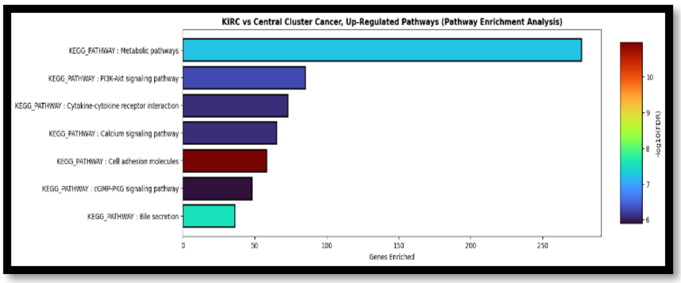

Figure 7

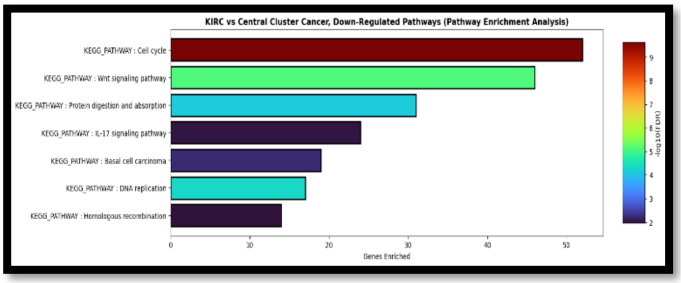

Figure 8

<u>Neural Cancers vs Central Cluster</u>

The upregulated pathways in the GBM + LGG cluster are dominated by neural signaling pathways such as neuroactive ligand-receptor interaction, GABAergic synapse, glutamatergic synapse, calcium signaling, and synaptic vesicle cycle (Fig. 9). This suggests that the separation of these cancer types into a distinct cluster is not necessarily due to a different mechanism of oncogenesis per se, but rather due to differences in cell-lineage.

Recent literature suggests that glioma cells can receive glutamatergic synaptic input from neurons, and that these neuron-glioma electrical interactions can activate downstream signals that support glioma growth and proliferation. [11]

Therefore, the upregulation of the glutamatergic synapse pathway may reflect a biologically relevant mechanism in glioma progression.

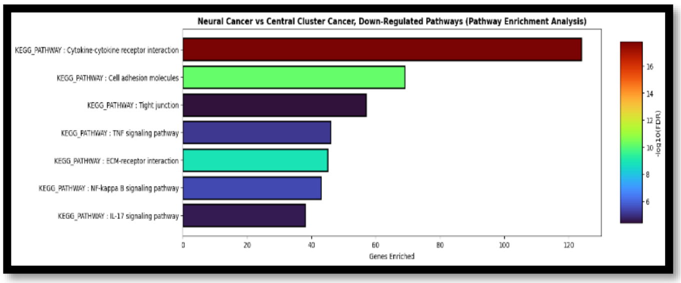

Figure 9

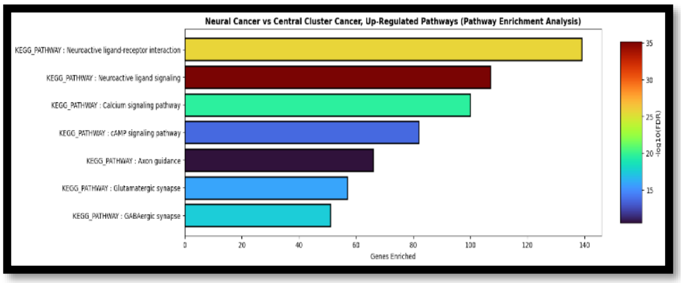

Figure 10

Additionally, the upregulation of gap junction-related pathways explains why neural cancers are difficult to treat, as these pathways are associated with tumor cell communication, coordinated survival responses and resistance to damage. Along with this, the blood-brain barrier further complicates treatment by limiting the ability of systemically administered drugs to effectively reach the tumor site.

There is a downregulation of DNA damage response pathways (p53 signaling pathway), apoptosis-related pathways such as NF-kappa B, IL-17 and TNF signaling pathways, along with motility-associated and cell cycle pathways (Fig. 10). This suggests that glioma progression in this comparison is not mainly driven by hyperproliferative mechanisms but instead, the pattern points more toward survival advantage and resistance to cell death.

THCA vs Central Cluster

The upregulated pathways in THCA consist predominantly of thyroid hormone synthesis and thyroid hormone signaling pathways (Fig. 11). The enrichment of these pathways reflects the thyroid-lineage of the cells, rather than being indicative of oncogenic mechanisms.

Additionally, the Ras-Raf-MEK-MAPK pathway is also seen to be upregulated. This is more directly relevant to thyroid cancer biology, as MAPK signaling is one of the major oncogenic pathways in thyroid carcinoma and is commonly activated through mutations involving BRAF, RAS etc. [12]

There is downregulation of inflammatory and apoptosis-related pathways such as cytokine-cytokine receptor interaction, IL-17 signaling, NF-kappa B signaling, TNF signaling, and chemokine signaling, along with DNA replication, p53 signaling, and cell-cycle-associated pathways (Fig. 12). Thus, THCA forms a distinct cluster not because of a completely unique oncogenic mechanism, but mainly because of its thyroid lineage-specific nature.

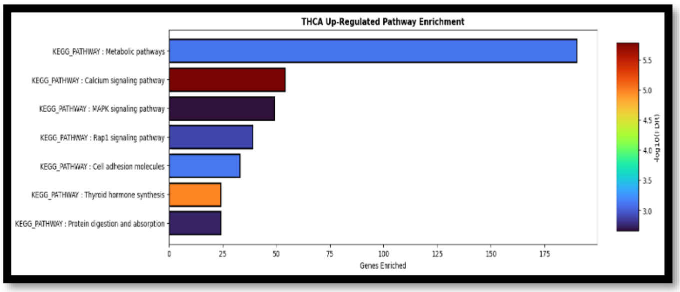

Figure 11

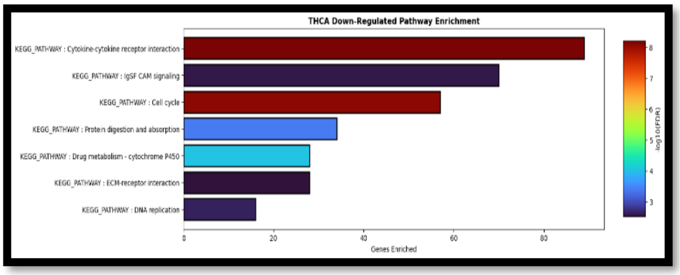

Figure 12

COAD + READ vs Central Cluster

The pathways upregulated in the cluster comprising COAD and READ (Fig. 13) include those associated with metabolism, uptake of bile acids, conversion of Retinoic Acid to Vitamin A and Steroid Biosynthesis.

Some of the genes upregulated in the Steroid Biosynthesis Pathway are responsible for the synthesis of Prostaglandins from their precursors.

Prostaglandins are known to inhibit apoptosis, induce angiogenesis and promote invasiveness in cancer cells, apart from maintaining an inflammatory micro-environment which promotes tumorigenesis. [13]

Furthermore, most of the genes upregulated in the Bile Secretion Pathway encode proteins that allow the transmembrane transport of bile acids.

Different Bile acids are known to either promote or suppress pathways associated with tumorigenesis such as apoptosis, angiogenesis, invasion, inflammation, etc. [14]

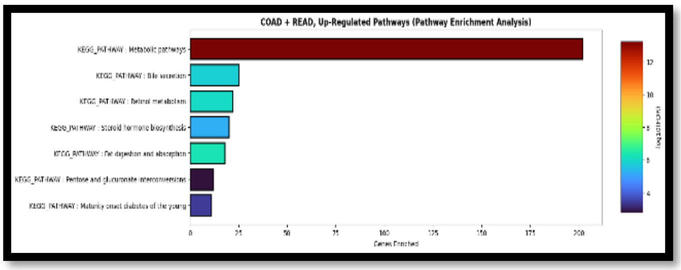

Figure 13

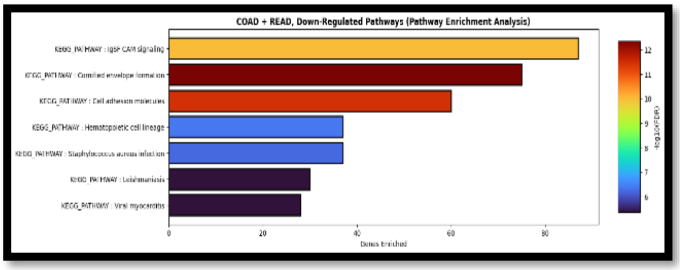

Figure 14

The pathways downregulated in COAD and READ (Fig. 14) are primarily associated with cell-ECM interactions; this observation agrees with the above inferences regarding the metastatic properties of these cancer cells.

For example, several genes associated with the downregulated pathways (shown above) encode different subunits of proteins such as claudins, integrins etc. that mediate interactions between cells and the ECM, reflecting the metastatic potential of the tumor cells.

Furthermore, several genes associated with the Hematopoietic Cell Lineage encode proteins that are known to promote differentiation, apoptosis, etc. These genes may have been downregulated in this case potentially due to their pleiotropic nature and given the characteristic loss of normal differentiation and inhibition of apoptosis observed in tumor cells.

Thus, we observe the modulation of certain pathways in colon and rectal glandular cells that promote the Epithelial-Mesenchymal Transition, as compared to other cancer cell types that constitute the central carcinoma cluster.

PRAD vs Central Cluster

The pathways upregulated in PRAD (Fig. 15) include those associated with metabolism and different signaling pathways.

In particular, the overexpression of genes encoding several proteins belonging to the cGMP-PKG signaling pathway, such as certain GPCRs, Akt, PKG, etc. is known to activate CREB, which can inhibit Caspase-3, and, therefore, apoptosis. This phenomenon has been observed in the case of several cancers including Acute Myeloid Leukemia, Lung Adenocarcinoma, etc. [15]

Several pathways that have been downregulated in PRAD (Fig. 16) are common with those downregulated in the case of COAD and READ.

These pathways consist not only those that mediate interactions between cells and the ECM but also those that maintain inter-cellular interactions (for example, some of the downregulated genes encode ICAMs).

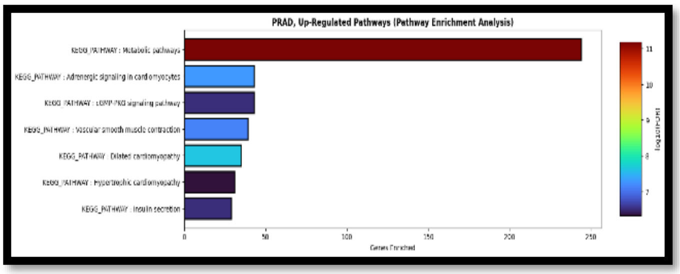

Figure 15

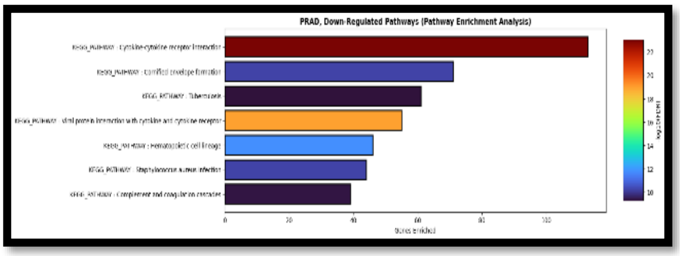

Figure 16

Downregulation of ICAM-1, in response to disruptive mutations of the RNA Endonuclease Dicer, which is known to inhibit micro-RNAs 222 and 339, has been known to reduce the susceptibility of tumor cells to recognition and lysis by Cytotoxic T-Lymphocytes (CTLs). [16]

Thus, we observe the deregulation of certain pathways in the case of prostate glandular cells that have been known to promote a tumor phenotype in different cell types, suggesting a broader role for these pathways in tumorigenesis.

Analyzing Similarity Between Tumor Types (using the Gaussian Naïve Bayes Algorithm)

As seen in Table 1 (Annexure), BRCA has the highest trace and determinant, varying the most across space, and has highly varying principal components. With a low condition number, it implies a large number of features are relevant to its heterogeneity.

KIRC and LUSC similarly have high determinants and traces, and comparatively small condition numbers. They follow a similar trend to BRCA.

On the other hand, UVM, READ and GBM have the lowest traces and small determinant values. Their data is more localized and less spread apart.

Reusing the model on the data displayed 100% accuracy. Thus, 769 components are over-surplus to our requirement of identifying similar tumors. Thus, the number of principal components were varied from 200 to 1, where the model started producing errors.

The last mismatched pair was LUAD and LUSC, both cancers of the lungs (Fig. 17). Given this is the last set to be separated, it suggests they are very similar. They have been noted to have similar signaling pathways (Notch, Hedgehog, Wnt and ErbB pathways) in literature, which would explain them being closely tied together. [17]

COAD and READ, CESC and LUSC, and BRCA and PRAD return a high number of misclassifications. The first pair is situated close to one another, and the second is of two squamous cell cancers. The last one corresponds to Breast and Prostate cancers, implicated together with both based on hormone receptors and linked with sex hormones. [18]

An interesting observation was the coupling between KIRC and THCA, two very distant tumors that were not expected to be similar. However, with similar gene expression profiles and increased lipid storages, they reduce accuracy in the confusion matrix. [19, 20]

Cancers prone to misdiagnosis are either spatially close, similar in terms of gene alterations, affected cell type or based on physiological changes. This indicates how tumors metastasize, share behaviors, and how often false positives or negatives occur during diagnosis.

Conclusion

From the above results, we see that most of the differences between types of cancer arise from their cell lineages and are rarely due to differences in oncogenic mechanisms.

Limitations

Since the data used is bulk RNA-Seq data, there is a likelihood that heterogeneities within cancer types might not have been captured.

Additionally, the dataset pools patients from different disease stages and includes cancers that may be specific to certain populations.

Future Directions

One could try to predict the patients pathological state using RNA-Seq data or incorporate that into the Differential Gene Expression Analysis to better understand tumor progression.

Accounting for relationships between genes, in very low dimensional space, could better represent the gene interaction network and aid in feasibly understanding the system.

Figure 17

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ANNEXURE

CANCER	TRACE	DETERMINANT	CONDITION NUMBER
BRCA	12329094.89	1e2404	1.55e5
CESC	3450144.29	-1e-4372	1.84e16
COAD	2859811.97	-1e-2614	1.01e16
ESCA	2867431.35	-1e-6137	3.54e17
GBM	1720808.71	-1e-6739	7.38e15
KIRC	7034349.78	-1e-1346	-9.55e15
LGG	5518358.59	-1e-1696	-3.55e16
LUAD	5608165.63	-1e-1367	-2.21e17
LUSC	6496030.89	-1e-1487	-5.35e16
OV	3609977.96	-1e-4345	1.18e16
PRAD	3322957.35	-1e-1954	1.54e16
READ	1160084.36	-1e-6826	4.57e15
TGCT	4202244.35	-1e-6876	-6.14e15
THCA	2919410.22	-1e-2025	-8.24e16
THYM	2261207.19	-1e-7219	-1.58e16
UVM	1255232.03	-1e-7933	-1.71e16

Table 1

Link to Google Drive (Supplementary):
https://drive.google.com/drive/folders/1maL6g2Uo9JizLPmGVTD3gTLns5z8lC0X